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Oefeningenreeks 3A nr. 39.3

$$f(x) = x^{2x+1}$$

$$f(x) = e^{(2x+1)\ln x}$$

$$f'(x) = e^{(2x+1)\ln x}((2x+1)\ln x)'$$

$$= (2\ln x + (2x+1)x^{-1})$$

$$f'(x) = x^{2x+1}(2\ln x + (2x+1)x^{-1})$$

$$= x^{2x+1}(2\ln x + 2 + x^{-1})$$

$$= x^{2x+1}(2x + 1 + 2x\ln x)x^{-1}$$

$$= x^{2x}(2x + 1 + 2x\ln x)$$

$$\boxed{f'(x) = x^{2x}(2x + 1 + 2x\ln x)}$$

References